BEGINNER LEVEL — Build the Foundations

Goal: Become fluent in core ML concepts and essential math.

1. Programming & Tools

Python fundamentals

NumPy, Pandas, Matplotlib

Jupyter, Git, virtual environments

2. Math for AI

Linear algebra (vectors, matrices, eigenvalues)

Probability basics

Calculus for optimization

Statistics (mean, variance, distributions)

3. Classical Machine Learning

Supervised learning: regression, classification

Unsupervised learning: clustering, PCA

Model evaluation: cross-validation, metrics

Overfitting, regularization

4. Essential ML Libraries

scikit-learn

XGBoost / LightGBM

Beginner Projects

Titanic survival prediction

MNIST digit classifier

Customer segmentation (K-means)

INTERMEDIATE LEVEL — Build Real AI Systems

Goal: Become a capable AI engineer who can train, deploy, and evaluate models.

1. Deep Learning

Neural networks (MLP, CNN, RNN)

Optimization (Adam, LR scheduling)

Regularization (dropout, batch norm)

Transfer learning

2. NLP Foundations

Tokenization, embeddings

RNNs, LSTMs, seq2seq

Attention mechanism

Transformers (BERT, GPT-style models)

3. Computer Vision

CNN architectures (ResNet, EfficientNet)

Object detection (YOLO, Faster R-CNN)

Segmentation (U-Net)

4. MLOps & Deployment

FastAPI, Docker

Model serving patterns

Experiment tracking (MLflow, W&B)

Monitoring & drift detection

5. Data Engineering for AI

ETL pipelines

Spark, Kafka basics

Data quality & governance

Intermediate Projects

Image classifier with transfer learning

BERT-based text classifier

End-to-end ML pipeline with FastAPI + Docker

Real-time inference with Kafka

ADVANCED LEVEL — Research-Grade & Industry-Grade AI

Goal: Operate at the level expected in R&D labs, PhD programs, and advanced AI engineering.

1. Advanced Deep Learning

Large-scale training

Distributed training (DDP, DeepSpeed)

Optimization at scale

Model compression & quantization

2. Large Language Models (LLMs)

Transformer internals

Pretraining vs. fine-tuning

LoRA, QLoRA, PEFT

Safety, alignment, RLHF

Evaluation frameworks

3. Retrieval-Augmented Generation (RAG)

Embeddings & vector databases

Chunking strategies

Query transformation

Hybrid search

Evaluation of RAG systems

4. Multi-Agent Systems

Agent architectures

Planning & tool-use

Memory systems

Coordination & communication

Multi-agent evaluation

5. Probabilistic Modeling

Bayesian inference

Probabilistic programming (Pyro, TFP)

Uncertainty quantification

6. Responsible & Trustworthy AI

Explainability (SHAP, LIME)

Fairness metrics

Robustness testing

Privacy-preserving ML (DP, FL)

Advanced Projects

Build a RAG system with evaluation

Fine-tune an LLM using LoRA

Multi-agent workflow orchestrator

Distributed training of a custom model

Bayesian optimization for hyperparameters